

Problem-Solution Fit

Date	15 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Use the framework below to ensure the proposed solution accurately addresses the root causes, behaviors, and limitations of the target audience.

1. CUSTOMER SEGMENT(S) [CS] - Residents/farmers in monsoon-prone river basin villages - Local disaster management officers coordinating regional flood response	6. CUSTOMER LIMITATIONS [CL] <i>eg. budget, devices</i> - Limited or intermittent internet connectivity in rural areas - Low-cost smartphones / basic browsers only - No budget for paid weather/telemetry services	5. AVAILABLE SOLUTIONS [AS] <i>pros & cons</i> - Govt. meteorological bulletins — authoritative but state-wide, not hyperlocal - Physical river gauge telemetry — accurate but too costly for small communities - Community word-of-mouth — free and fast but unreliable
2. PROBLEMS / PAINS [PR] <i>+ its frequency</i> - No advance, hyperlocal flood warning — recurring every monsoon season (Jun–Sep, annually) - Forecasts are regional, not specific to one's own village or basin	9. PROBLEM ROOT / CAUSE [RC] No affordable, accessible tool converts raw seasonal rainfall/cloud-cover data into an instant, localized flood-risk score	7. BEHAVIOR [BE] <i>+ its intensity</i> - Checks local news/social media multiple times a day during monsoon (high intensity, seasonal) - Pre-emptively moves valuables and grain to higher ground every year
3. TRIGGERS TO ACT [TR] - Onset of monsoon season - Heavy overnight rainfall or news of upstream flooding - Community elder's warning	10. YOUR SOLUTION [SL] Rising Waters: a free, lightweight web app — enter 5 rainfall/cloud-cover readings and get an instant ML-predicted flood-risk verdict with a confidence score, usable on any device with a browser	8. CHANNELS OF BEHAVIOR [CH] ONLINE Local news sites, WhatsApp/community groups, search engines
4. EMOTIONS [EM] <i>before / after</i> - Before: anxious, uncertain, unprepared each monsoon - After: informed and confident (safe result) or alert and prepared (flood-risk result)		OFFLINE Community announcements, word of mouth, local radio